MATH MA’AM ACADEMY

Where Every Subject Clicks.

STD 5 MATHEMATICS – CHAPTER 7

SYMMETRY

Learning Objective: Understand symmetry and identify the line(s) of symmetry in common geometrical shapes and familiar figures.

7.1 WHAT IS SYMMETRY?

Symmetry is a property of a shape or object in which one half is the mirror image of the other half. When the two halves match exactly after folding or reflection, the object is symmetrical.

Line of Symmetry

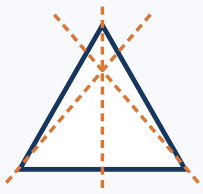
A **line of symmetry** is an imaginary line that divides a shape into two equal, matching mirror-image halves. It is also called a **mirror line** or **axis of symmetry**.

Remember: The two halves must match exactly in shape and size. Merely dividing a figure into two equal areas does not always make the dividing line a line of symmetry.

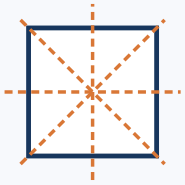
How to identify a line of symmetry

1. Imagine folding the shape along the proposed line.
2. Check whether one half covers the other half completely.
3. If the edges and corners coincide exactly, the fold is a line of symmetry.
4. A shape may have one, more than one, or no line of symmetry.

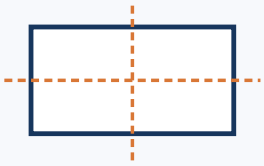
Examples of symmetry



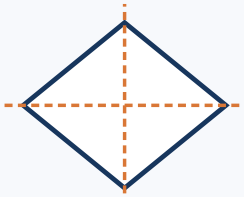
Equilateral triangle



Square



Non-square rectangle



Rhombus

Number of lines of symmetry in common shapes

Shape	Number of lines	Important point
Equilateral triangle	3	One line from each vertex to the midpoint of the opposite side.
Isosceles triangle (not equilateral)	1	The line passes through the top vertex and midpoint of the base.
Scalene triangle	0	All sides are different; no mirror line.
Square	4	Two midlines and two diagonals.
Non-square rectangle	2	One vertical and one horizontal centre line.

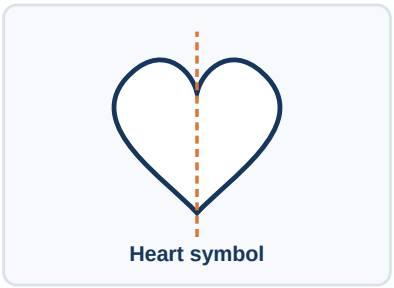
Shape	Number of lines	Important point
Rhombus (not square)	2	Its two diagonals are symmetry lines.
Circle	Infinitely many	Every diameter is a line of symmetry.
Semicircle	1	The line passes through the centre and is perpendicular to the diameter.

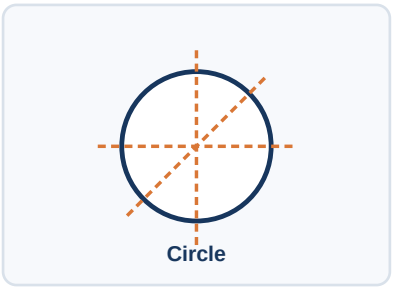
SYMMETRY IN FIGURES AND LETTERS

Symmetry in everyday figures

Many objects around us show symmetry. Examples include butterflies, some flowers, leaves, the front view of many buildings, and some road signs. The two sides of a butterfly's body are approximately mirror images.

Butterfly


Heart symbol


Circle

Symmetry in capital letters

In standard block-style English capital letters, some letters have line symmetry. The exact appearance can depend on the font used.

Letter	Line(s) of symmetry	Description
A	1 vertical	Left and right slanting sides match.
H	2 (vertical and horizontal)	Both centre lines divide it into matching halves.
I	Usually 2	In a simple block font with equal top and bottom bars.
O	Many / infinitely many if circular	A geometrically circular O is symmetric about every diameter.
F	0	It has no mirror line in the usual block style.

Reflection and mirror images

Reflection means flipping a figure over a line to make its mirror image. The line used for the flip is the mirror line or line of symmetry.

Properties of a reflection:
- The image has the same shape and size as the original figure.
- Corresponding points lie on opposite sides of the mirror line.
- Each point and its image are at equal perpendicular distances from the mirror line.
- A figure reflected across its own line of symmetry fits exactly on the original figure.

Symmetry in grids

When a pattern is drawn on a square grid, compare each shaded square with the square on the opposite side of the mirror line. The matching square must be the same distance from the line, measured at right angles to it. A square on the mirror line remains in the same position.

Grid reflection idea

Important facts to remember

- A line of symmetry divides a figure into two mirror-image halves.
- A shape can have zero, one, two, three, four or many lines of symmetry.
- A square has 4 lines; a non-square rectangle has 2; an equilateral triangle has 3.
- A scalene triangle has no line of symmetry.

- Every diameter of a circle is a line of symmetry.
- A regular polygon has the same number of lines of symmetry as it has sides.

QUICK REVISION & PRACTICE

Common mistakes

1. Do not count a line just because it divides a shape into two parts. The two parts must be mirror images.
2. Do not confuse a diagonal with a symmetry line. A diagonal is a line joining opposite vertices; it is a symmetry line only when the two sides match.
3. A non-square rectangle has 2 symmetry lines, not 4. Its diagonals are not symmetry lines.
4. A circle has infinitely many symmetry lines, not just the vertical and horizontal ones shown in a drawing.

Solved examples

Example 1: How many lines of symmetry does a square have?
Answer: 4 — two lines through the midpoints of opposite sides and two diagonals.

Example 2: A triangle has two equal sides but its third side is different. How many lines of symmetry does it have?
Answer: 1 — the line from the vertex between the equal sides to the midpoint of the base.

Example 3: Does a general slanting parallelogram have a line of symmetry?
Answer: No. A parallelogram that is not a rectangle or rhombus has no line of symmetry.

Practice questions

1. Define symmetry in your own words.

2. What is a line of symmetry? Write one other name for it.

3. How many lines of symmetry does each have: (a) square (b) equilateral triangle (c) scalene triangle (d) circle?

4. Draw a rectangle and mark all its lines of symmetry.

5. Draw an isosceles triangle and show its line of symmetry.

6. Why are the diagonals of a non-square rectangle not lines of symmetry?

7. What happens when a shape is folded along its line of symmetry?

8. A point is 4 cm from a mirror line. How far is its reflected image from the mirror line?

9. Name two objects from everyday life that show approximate symmetry.

10. In a square grid, what should you check to decide whether two shaded squares are mirror images?

> **Exam tip:** Practise drawing the symmetry lines yourself. Use a ruler for straight axes and fold paper shapes to check whether the two halves overlap exactly.

ALL THE BEST